FINAL VIVA · ANLP MONSOON 2026

C3: Committed Confidence Calibration

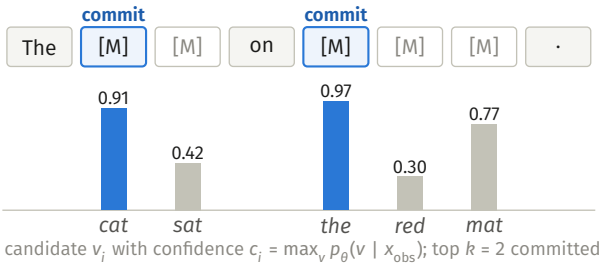
Selection bias and two sources of error in confidence-based token commitment in diffusion language models

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Diffusion LMs commit the tokens they are most confident about

One forward pass scores every masked position; the decoder keeps the top-k



- LLaDA-8B and Dream-7B keep the most confident positions (MaskGIT-style); Fast-dLLM puts one global threshold on the same number.
- A committed token won: its confidence was the largest of m noisy estimates. That is the setting of the winner's curse (Efron 2011; Andrews et al. 2024).

Question. Once a token has won, is its confidence still an honest probability of being right? And when several tokens win in one step, which error dominates?

Two separate ways parallel decoding goes wrong

For a set S committed in one step, given the observed context x_{obs}

$$\mathrm{KL}\Big(p^* \,\Big\|\, \prod_{i\in S} p_{\theta,i}\Big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i\in S} \mathrm{KL}\big(p_i^* \,\|\, p_{\theta,i}\big)}_{\text{(B) estimation}}$$

- **(A)** Marginals, never the joint: tokens can each be plausible yet jointly nonsensical.
- **(B)** The marginals themselves are wrong. The dependency literature treats **(B)** as fixed.
- C2 measures both where p^* is known; C1 audits calibration under selection.

Work	Calibrated under selection?	Separates (A) and (B)?
Fast-dLLM; OSDT	no	no
KLASS; SWD; LESS	no	no
PC-Sampler	positional only	no
DAPD; DEMASK; PUNT	no	(A) only
ParallelBench	no	(A) only
Tang et al. (2026)	no	identifies
This work	**yes**	**both**

Three contributions

C1

Selection-conditional calibration audit

Is a token that *won* its pool overconfident? On LLaDA-8B, Dream-7B (Base and Instruct) and a 110M masked diffusion LM we trained with our evaluation documents held out. Numbers from a held-out report half.

C2

Both terms of the KL split

Exactly, in nats, on a synthetic JSON probe where p^* is known. On real text, where it is not, proxies whose behaviour we first check on the probe: a direction, not a split in nats.

C3

A recalibrated commit rule

A threshold on recalibrated confidence, built only from what the decoder already computes (no extra forward passes), against six decoders at matched tokens per step on GSM8K and MBPP.

Order of the talk. Theory and the instrument, then C1, C2 and C3 in turn, then what we got wrong and what remains open.

Theory: where a winner's curse can and cannot come from

State s: a document with masked set M. Teacher forcing makes correctness $a_i = 1\{v_i = x_{0,i}\}$ observable

Proposition 1 · Winning adds nothing given confidence and context

If the pairs (ℓ_j, a_j) are independent across $j \in M$ given s, and the pool P is drawn independently of them, the winner i of P satisfies

$$\Pr\left(a_i = 1 \mid \ell_i,\ s,\ i = \arg\max_{j \in P} \ell_j\right) = \pi(\ell_i, s).$$

However unequal the noise scales: heteroskedastic noise alone cannot bias committed confidence. This corrected the proposal's argument.

Proposition 2 · Exact split of the committed-token gap

$$\Delta(m) = \underbrace{\mathbb{E}_m[\sigma(\ell) - \pi(\ell)]}_{\text{own miscalibration}} + \underbrace{\mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]}_{\text{field term}}$$

$\pi(\ell)$: ordinary calibration curve; $\pi(\ell, s)$: its context-aware version. With a pool of one, winners are random, so the field term is exactly 0.

The field term is nonzero only when accuracy at fixed confidence depends on context: the **winner's curse in a corrected, testable form**. Proofs in the backup.

C1: what we measure, and when we declare an effect

Sign: confidence − accuracy, so positive means overconfident (the proposal used the opposite)

E1 · committed-token gap

$\Delta(m) = \mathbb{E}_m[c_i - a_i]$ over tokens that win pools of size $m \in \{1, 2, 4, 8, 16, 32\}$. Primary contrast $\Delta(32) - \Delta(1)$.

E2 · field effect

Observed $\Delta(m)$ minus the gap predicted by $\hat{\pi}(\ell)$, which sees only the token's own confidence. A context-aware $\hat{\pi}(\ell, s)$ should cover the rest.

E3 · trajectory shift

Training uses uniform masks; decoders fill easy positions first. Are decoder states less calibrated at equal confidence and matched mask rate?

Hypotheses. **H1** $\Delta(32) - \Delta(1) > 0$; **H2** the field term at $m = 32$ is positive; **H3** decoder states are more overconfident than uniform ones.

Declare rule. The 95% interval excludes zero *and* the estimate exceeds twice the non-determinism floor. The floor is 0 for all three model families, so $\delta_{\min} = 0.02$, fixed in advance, sets power. Holm across models × datasets.

Diagnostics fitted on winning tokens:

$$\Pr(a_i{=}1) = \sigma\big(\alpha\,[+\alpha_s] + f(\ell_i) + \beta_z z_{r,m}\big),$$

or rank dummies γ_k in place of $z_{r,m}$; f a 9-knot spline fixed in advance. State effects α_s absorb the field term: $\beta_z < 0$ without them and ≈ 0 with them signals a field effect.

C1: the trace sweep

Collect once on GPU, one row per (masked state, position); draw pools and fit on CPU

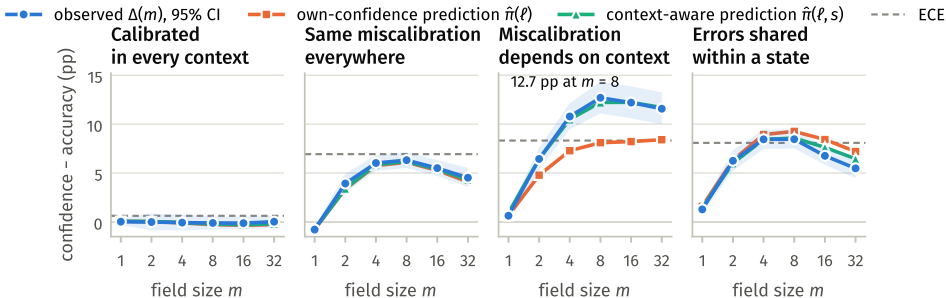
Model	Text	Prec.	Docs	Rows (M)	GPU-h
LLaDA-8B Base	FineWeb-Edu	fp16	1,979	119.6	68
Dream-7B Base	FineWeb-Edu	fp16	1,976	119.4	60
Our 110M MDLM	FineWeb-Edu	fp32	1,979	119.6	3
LLaDA-8B Instruct	GSM8K	fp16	1,319	13.8	18
LLaDA-8B Instruct	MBPP	fp16	499	3.1	5
Dream-7B Instruct	GSM8K	fp16	1,319	13.4	14
Dream-7B Instruct	MBPP	fp16	499	2.5	4
LLaDA-8B Base	time-split	fp16	500	30.2	17
Dream-7B Base	time-split	fp16	500	30.2	15
Our 110M MDLM	time-split	fp32	500	30.2	1
LLaDA-8B Base	FineWeb-Edu	4-bit	198	12.0	5

Top: the primary arms. Grey: the unseen time-split set and the 4-bit control (T8). RTX 2080 Ti: 208 GPU-hours, 0.49 billion rows.

- **States per document**: uniform masks at rates 0.1–0.9; teacher-forced confidence top-k trajectories, $k \in \{2, 4, 8\}$; uniform masks matched to them (466 passes per document).
- **Per row**: candidate, confidence, logit, correctness, p(true), entropy, margin, top-20.
- **Halves**: 1,000 fit / 1,000 report documents. Calibrators, splines and knots from the fit half; every C1 number from the report half.
- **Deterministic**: repeats are bitwise identical, so the floor is 0 for every family.

The instrument recovers known truth (T0)

Four simulated worlds · fit half → report half · document bootstrap



Why ECE is not enough. When miscalibration depends on context, ECE is 8.3 pp but Δ(32) − Δ(1) = 10.9 pp. Known weakness: the field-term estimator reads high (3.3 pp against a true 1.6 pp).

C1 results: does winning make a token overconfident? (E1, E2)

Every arm, in uniform states and in the states of the teacher-forced top-k decoder (k=4)

	E1 = Δ(32) − Δ(1)		E2 = field contrast	
Arm	uniform	decoder	uniform	decoder
LLaDA Base	−0.003*†	0.006*†	0.001*†	0.004*†
Dream Base	−0.002*†	0.015*†	0.001*†	0.003*†
LLaDA-I GSM8K	−0.022*†	0.024*†	−0.003*†	0.021*†
LLaDA-I MBPP	0.025†	0.156*†	0.013*†	0.036*†
Dream-I GSM8K	0.061*†	0.072*†	0.021*†	0.038*†
Dream-I MBPP	0.257*†	0.186*†	0.176*†	0.076*†
MDLM-110M	0.002†	0.009*†	0.002*†	−0.002*†

E1 = Δ(32) − Δ(1). Positive: a token that won a field of 32 is more overconfident than a random masked position.

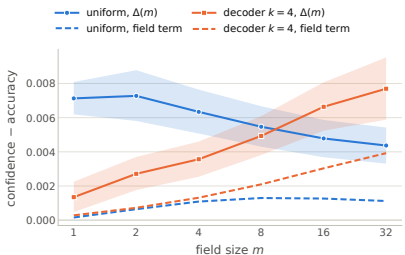
E2 = field(32)−field(1). Positive: part of that growth is the winner's curse proper (Proposition 2).

Markers. * Holm-rejected in its family; † declared (95% interval excludes 0, effect above twice the floor). **A claim needs both.**

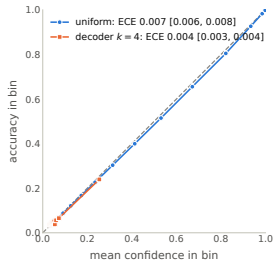
> **Claims.** E1: 12 of 14 cells. E2: 14 of 14. E1 is positive in every decoder-state cell; E2 clearly exceeds the estimator's T0 bias only for the instruct models.

C1 results: how the gap grows with the field, and how much is the field term

Main arm: LLaDA-8B-Base; the other arms are in the backup



Left. Blue: uniform; orange: decoder states. Solid $\Delta(m)$, dashed its field term; the gap between them is own miscalibration. Rising solid: E1; rising dashed: E2.



Right. Below the diagonal: confidence exceeds accuracy. ECE sees only this picture, not the split into own and field terms (T0).

C1 diagnostic: does winning a larger field predict correctness? (β_z)

$\Pr(a_i{=}1) = \sigma(\alpha\,[+\alpha_s] + f(\ell_i) + \beta_z z_{r,m})$, without and with state effects α_s

	β_z, no state effects		β_z, state effects	
Arm	uniform	decoder	uniform	decoder
LLaDA Base	—	—	—	—
Dream Base	—	—	—	—
LLaDA-I GSM8K	—	—	—	—
LLaDA-I MBPP	—	—	—	—
Dream-I GSM8K	—	—	—	—
Dream-I MBPP	—	—	—	—
MDLM-110M	—	—	—	—

Field effect. $\beta_z < 0$ without state effects and ≈ 0 with them; it should agree with a positive E2.

Shared errors. $\beta_z \neq 0$ with state effects: positions of a state are wrong together, Proposition 1 fails, and E2 is not a field effect.

Caution. In T0, shared errors gave a *positive* β_z without state effects and a negative field term, so we read β_z only beside the sign of E2.

> A diagnostic: no Holm family, no decision. Point estimates here; intervals and convergence flags are in the backup.

C1 results: are the states a decoder visits less calibrated? (E3)

Decoder-state tokens against uniform states with as many masks, at equal confidence

RESULTS LAND TUE 27 OCT

Waits for `paper/figures/c1/c1_e3_LladaBase.pdf`, written by `scripts/c1_paper_outputs.py` from `results/e3/LladaBase/results.json` (`scripts/analyze_e3.py`).

Left: reliability of decoder states against matched-uniform states, with bin intervals. Right: the accuracy difference against confidence, with pointwise and simultaneous 95% bands. Main arm LLaDA-8B-Base, k=4.

Right panel. Type effect > 0: at equal confidence decoder-state tokens are right less often. Teacher-forced: this is which positions remain masked, not the decoder's own mistakes. All E3 rows are exploratory.

k=4, matched uniform: type effect ? [?, ?], sup-t p = ?.

C1 controls: do position, token type or temperature explain E1, E2? (T8)

Each control removes one alternative explanation of the growth with m

	E1 = Δ(32) − Δ(1)				E2 = field contrast		
LLaDA-8B-Base	plain	+ position	+ token type	+ temperature	plain	+ position	+ token type
uniform	—	—	—	—	—	—	—
decoder k=4	—	—	—	—	—	—	—

4-bit against fp16 (198 documents): E1 changes by ? [?, ?] in uniform and ? [?, ?] in decoder states.

How to read. A column close to plain: the control did not matter; near 0: the growth was where or what winners are. ‡: the paired with-minus-without 95% interval excludes 0.

C2: a JSON probe where the true joint p^* is known

Our 110M model, fine-tuned on the probe; both terms computed exactly at every decoding step

- **Instances**: the simple fields of 200 SchemaStore schemas, 6–16 per schema, single-token values with seeded Dirichlet marginals.
- **Dependency knob** ρ: fields are paired. With probability ρ, stated in a marker at the start, the follower is a fixed function of its leader; otherwise an independent draw. So p^* is exact.
- **Model**: our 110M fine-tuned on 200k instances; evaluated on 1,000 (200 per ρ).
- **Decoding**: all values masked; each step a rule picks k fields, the most confident or at random, and we fill them with the true values.
- **Check**: **(A)** + **(B)** equals the directly computed KL; largest error 4.3×10^{-14} nats (bar 10^{-6}).

one evaluation instance (ρ = 1), values masked

```
{"_rho":"1.00",
  "bundle":[M], "path":[M],
  "ref":[M], "repo":[M],
  "stats":[M], "version":[M]}
```

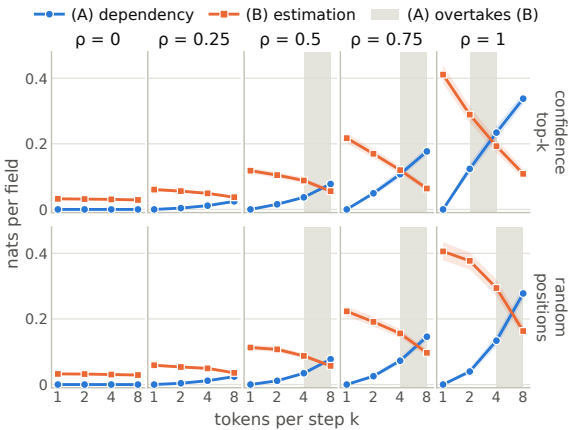
Per step, for the committed set S

(A) = $\sum_{i \in S} H(p_i^*) - H(p_S^*)$: zero unless a leader and its follower are committed together.

(B) = $\sum_{i \in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})$: mass off the allowed values counts against **(B)**.

Summed over steps, per field, bootstrapped over instances.

C2: dependency overtakes estimation only at high ρ and k



- **(A)** is zero at $k = 1$ and at $\rho = 0$, and grows with both.
- For $\rho \geq 0.5$, **(A)** overtakes **(B)** at $k = 8$; at $\rho = 1$ under confidence top-k already at $k = 4$. Each crossover is also where the 95% interval of **(A)** –**(B)** clears zero.
- For $\rho \leq 0.25$, **(B)** dominates up to $k = 8$.

$\rho = 1$, $k = 8$, confidence top-k

(A) 0.338 [0.320, 0.355] and **(B)** 0.109 [0.095, 0.123] nats per field: **(A)** is 76% of the total.

Main model: the fine-tune that never saw the ρ marker masked (choice in the backup). 10,000 bootstrap resamples over instances.

C2: three findings that correct our plan

1 · Confidence selection raises (A), not (B)

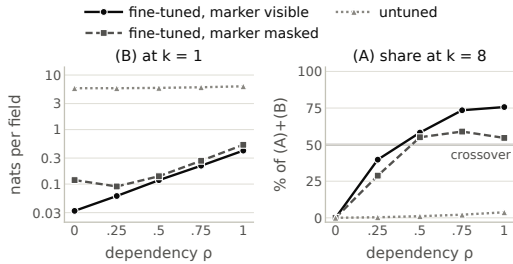
Planned: ranking on noisy confidences inflates **(B)**.
Where **(B)** is higher than at random, the intervals overlap.
(A) rises instead (ρ = 1, k = 2): 0.124 [0.108, 0.140] against 0.040 [0.031, 0.049] at random.

2 · (B) depends on ρ

Planned: ρ moves only **(A)**.
At k = 1, **(B)** rises from 0.032 at ρ = 0 to 0.411 at ρ = 1: the model reads the marker only in part.

3 · (A) and (B) trade off along k

At ρ = 1, from k = 1 to 8, **(B)** falls from 0.411 to 0.109 while **(A)** rises from 0 to 0.338: a follower decoded with its leader shows its error as **(A)**, not **(B)**.



- Without fine-tuning, **(B)** dominates everywhere: **(A)** is at most 4% of the total.
- The ranking commits leaders together with their followers more often than chance, which brings the crossover forward.

C2 on real text: the dependency proxy is near zero

300 fit-half FineWeb-Edu documents, one state per mask rate; the proxy is first checked on the probe

Proxies

For **(A)**: **parallel–sequential disagreement**, the fraction of the k most confident positions whose token changes when committed one at a time rather than in one pass. For **(B)**: the gap $\Delta = c - a$ and the cross-entropy (CE) of the true token.

On the probe, it tracks (A) weakly

Spearman with the exact **(A)** over the 15 confidence cells with $k \geq 2$: 0.49 [0.28, 0.63]; 0.28 for the other fine-tune. It rises with k, barely with ρ: at $k = 8$, 0.028 at $\rho = 0$, where **(A)** is zero, and 0.049 at $\rho = 1$.

mask rates 0.1–0.9	110M (fp32)	LLaDA-8B Base (fp16)
disagreement, $k = 4$	0.0003	0.0001
disagreement, $k = 8$	0.0021	0.0022
CE (nats), $k = 1 \rightarrow 8$	0.08 → 0.40	0.08 → 0.23
Δ, $k = 1$	+0.001 [−0.003, 0.005]	+0.003 [−0.001, 0.006]
Δ, $k = 8$	+0.007 [0.004, 0.010]	+0.004 [0.002, 0.007]

CE includes the text's own entropy, and lower ranks are harder positions.

At 2–8 tokens per step, real text looks like the probe's **low-ρ regime, where (B) dominates**. But the proxy sees only dependence the model has learned: near-zero disagreement does not show that **(A)** is small.

C3a: let recalibrated confidence set how many tokens to commit

Built only from what the decoder already computes: no extra forward pass

- **Order**: the model's own confidence ℓ ranks the masked positions of the active block, as in top-k.
- **Count**: commit the top n, $n = \#\{\hat{c} \geq 1 - \alpha\}$ and at least one, with $\hat{c}$ the recalibrated confidence.
- **Calibrator**: context-aware logistic on decode-time field features (median ℓ, mask rate, $\log|M|$).
- **Fit once, then frozen**: fit-half LLaDA-8B-Instruct GSM8K and MBPP traces, decoder states $k \in \{2, 4, 8\}$ pooled; report halves never read.
- **Known shift**: free-running starts from 256 masks, fit states sit inside a reference answer; we count steps outside the fit range.

RESULTS LAND THU 29 OCT

Waits for the frozen calibrators in `artifacts/c3/` (`scripts/fit_c3a_calibrator.py`) and a macros file for their diagnostics.

- Development check: calibration error of $\hat{c}$ against raw $\sigma(\ell)$ on held-out fit-half documents (not a headline).
- Share of decode steps outside the fit range, per task.
- Share of fit states where $\hat{c}$ never decreases with ℓ.

C3: accuracy at matched tokens per step, against six decoders

LLaDA-8B-Instruct, free-running, 256 tokens in blocks of 32, greedy, report halves of GSM8K and MBPP

RESULTS LAND THU 29 OCT

Waits for `artifacts/t12/figures/t12_matched_speed.pdf` (proposed name), from the T12 scoring summaries (`scripts/score_t12.py`).

Accuracy against tokens per step, one panel per task, one curve per decoder at its three speeds, with 95% Wilson intervals.

Decoders. Top-k, Fast-dLLM threshold, entropy-bounded, margin, KLASS, PC-Sampler, and C3a, each at three speeds.

Reading. Up and to the right is better. Speed is matched by tokens per step, so compare accuracy at the same x.

Claim. C3a is a gain only if it is more accurate than the other decoders at the same tokens per step, beyond the intervals. Otherwise the answer is no gain at matched speed, and we say so.

> Answer in one line: *pending (T12)*

C3: does the harness reproduce known decoders, and at what cost?

Fewer steps need not mean less time: wall-clock and forward passes beside tokens per step

RESULTS LAND THU 29 OCT

Waits for `artifacts/t12/t12_sanity_cost.tex` (proposed name), from the T12 scoring summaries (`scripts/score_t12.py`) and the sanity run.

- Sanity row: top-k and the Fast-dLLM threshold on about 200 fit-half GSM8K problems at Fast-dLLM's settings (5-shot), against the published numbers; the plan's bar is 1–2 points.
- Per decoder and speed: tokens per step, forward passes per problem, wall-clock per problem, and the share of problems that failed to decode.
- C3a only: the share of steps outside the calibrator's fit range.

How to read. If the sanity row misses the published numbers, nothing after it is comparable and we say that first. A faster decoder in steps that is not faster in seconds is reported as such.

Summary

1 · Theory

Selection alone cannot bias a calibrated posterior (Proposition 1). Any bias lives in the field term (Proposition 2), which the instrument measures and which T0 shows it can see where ECE cannot.

2 · C1 audit

Claims: E1 12 of 14 cells, E2 14 of 14. In decoder states the gap grows with the field for every model; the field term grows clearly only for the instruct models.

3 · C2: (A) and (B) separated

On the probe, **(A)** overtakes **(B)** only for $\rho \geq 0.5$ and larger k; confidence selection raises **(A)**, not **(B)**. On real text at 2–8 tokens per step, the **(A)** proxy is near zero: the **(B)**-dominated regime.

4 · C3 decoding

Pending: C3a against six decoders at matched tokens per step (Thu 29).

What we got wrong, and the limits of these results

Claims we withdrew

- Heteroskedastic noise alone makes winners overconfident (proposal): false, Proposition 1.
- Selection on confidence inflates **(B)** (proposal, mid report): on the probe it raises **(A)** instead.
- ρ moves only **(A)** (plan v2): **(B)** rises with ρ too.
- C2 on an 8B model (plan v2): not run. The probe results are for our 110M only.
- Real-text C2 as identified contrasts (mid deck): they are proxies, with a direction but no split in nats.

Limits

- C2's exact values hold for one synthetic distribution and one 110M model.
- Neither fine-tune met the marker bar, so the main probe model is the one with lower validation NELBO. The other model's crossovers are in the backup.
- The disagreement proxy tracks **(A)** weakly, and mostly through k.
- 8B models in fp16, not bfloat16 (no bfloat16 on our GPUs); E3 for top-k decoders only.
- C1 is teacher-forced, and E2 intervals condition on the fitted calibrators.
- C3 limits: when T12 lands.

Thank you. Questions?

Code	github.com/abhirajratna/ANLP-Project
Weights & Biases	wandb.ai/ANLP-project/c3-dllm-calibration
Hugging Face	huggingface.co/ANLP-project
Paper	DuckDuckDuckGoose-Final.zip

Backup: proofs

W_i: the event that $i \in P$ wins, $i = \arg\max_{j \in P} \ell_j$; ties broken by independent randomness U

Proposition 1

Suppose, given s, each pair (ℓ_i, a_i) is independent of $\ell_{-i} = (\ell_j)_{j \in M \setminus \{i\}}$. Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$.

Given s, $(\ell_i, a_i) \perp \ell_{-i}$ and $(P, U) \perp (\ell_j, a_j)_{j \in M}$, so $(\ell_i, a_i) \perp Y \mid s$, and by weak union $a_i \perp Y \mid Z$, i.e. $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$.

W_i is determined by (Y, Z), so for every bounded measurable h

$$\mathbb{E}[a_i h(Z) 1_{W_i}] = \mathbb{E}[\pi(\ell_i, s) h(Z) 1_{W_i}].$$

Dividing by $\Pr(W_i) > 0$ gives $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s.

Proposition 2

Exactly one W_i holds. Applying the display with $h \equiv 1$, summing over i and averaging over s:

$$\mathbb{E}_m[a] = \sum_i \mathbb{E}_m[a_i 1_{W_i}] = \sum_i \mathbb{E}_m[\pi(\ell_i, s) 1_{W_i}] = \mathbb{E}_m[\pi(\ell, s)],$$

so $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$; add and subtract $\pi(\ell)$.

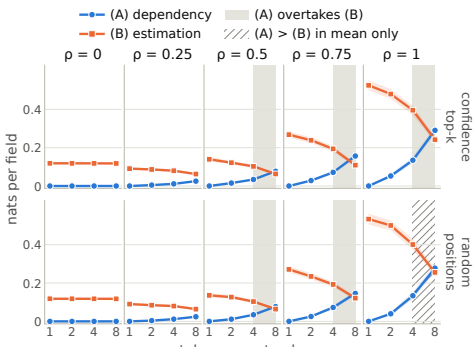
At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ by the display, so the field term is 0.

Backup: how the main probe model was chosen

Recorded in `artifacts/probe/main_variant.json`

	ft110m-v3-visible	ft110m-v3
ρ marker in fine-tuning	never masked	masked
Fine-tuning tokens (M)	300	100
Marker effect on copy	0.251	0.014
Val. NELBO (nats/tok.)	0.1751	0.1892
First k with **(A)** > **(B)** (ρ=1)	4	8

- Neither model reaches the marker bar of 0.5, so the main model is the one with the lower validation NELBO.
- The other model crosses over at k = 8 for every $\rho \geq 0.5$ under both rules: the qualitative picture holds.



As the main figure, for ft110m-v3. Hatched: **(A)** > **(B)** in mean only.

Backup: disagreement by mask rate (T10)

Disagreement at $k = 8$ per mask rate; one state per document per rate

Mask rate	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
110M	0.0004	0	0	0	0	0	0	0.0013	0.0171	0.6250
LLaDA-8B-Base	0	0	0	0	0	0	0	0.0008	0.0192	0

- At rate 1 every input is the same all-`[MASK]` sequence. That one state gives the 110M model nearly all of its disagreement (0.0644 at $k = 8$ over all ten rates); LLaDA does not disagree there. The main slide reads rates 0.1–0.9.
- Documents under 256 tokens are skipped; at least 8 masked positions per state; 10,000 document-bootstrap resamples.

Backup: data, contamination and our 110M model

- **2,000 calibration documents**, reservoir-sampled from FineWeb-Edu with a fixed seed and excluded from our 110M's training stream; 1,000 fit / 1,000 report.
- **Tasks**: GSM8K and MBPP test sets, split 50/50 the same way, as teacher-forced documents.
- **Contamination screen (T4)**: prompt with 50 tokens, continue greedily, flag a verbatim run ≥ 32 tokens. Flagged: 2/1,979 (LLaDA), 1/1,976 (Dream); the 8B arms are also analysed without them.
- **Time-split set**: text written after the 8B models were trained.

Model	bidirectional transformer: 12 layers, width 768, 12 heads; sequence length 256
Objective	MDLM (weighted masked cross-entropy), GPT-2 BPE + [MASK]
Training	169,055 steps, 10.0B tokens of FineWeb-Edu

37.2

perplexity *bound* on the 2,000 held-out docs [36.5, 38.0]

28.5

GPT-2 small, *exact* perplexity, same docs [28.0, 29.0]

MDLM gives only an upper bound, and GPT-2 saw other data with far more compute.

Backup: C1 primary endpoints, every arm, with intervals

95% intervals and the Holm decision in the 14-test families; TS: time-split, outside them

Arm	Masks	E1	Holm	E2	Holm	β_z	β_z^{FE}
LLaDA Base	uniform	−0.003*,† [−0.004, −0.002]	yes	0.0010*,† [0.0009, 0.0010]	yes	—	—
	dec. k=4	0.006*,† [0.005, 0.008]	yes	0.0036*,† [0.0035, 0.0037]	yes	—	—
Dream Base	uniform	−0.002*,† [−0.004, −0.001]	yes	0.0011*,† [0.0011, 0.0012]	yes	—	—
	dec. k=4	0.015*,† [0.013, 0.017]	yes	0.0029*,† [0.0028, 0.0029]	yes	—	—
LLaDA-I GSM8K	uniform	−0.022*,† [−0.025, −0.018]	yes	−0.003*,† [−0.004, −0.003]	yes	—	—
	dec. k=4	0.024*,† [0.016, 0.032]	yes	0.021*,† [0.020, 0.023]	yes	—	—
LLaDA-I MBPP	uniform	0.025† [0.003, 0.051]	no	0.013*,† [0.010, 0.016]	yes	—	—
	dec. k=4	0.156*,† [0.126, 0.189]	yes	0.036*,† [0.031, 0.041]	yes	—	—
Dream-I GSM8K	uniform	0.061*,† [0.052, 0.070]	yes	0.021*,† [0.020, 0.023]	yes	—	—
	dec. k=4	0.072*,† [0.064, 0.081]	yes	0.038*,† [0.036, 0.039]	yes	—	—
Dream-I MBPP	uniform	0.257*,† [0.227, 0.290]	yes	0.176*,† [0.159, 0.194]	yes	—	—
	dec. k=4	0.186*,† [0.151, 0.224]	yes	0.076*,† [0.066, 0.086]	yes	—	—
MDLM-110M	uniform	0.00152† [0.00003, 0.00301]	no	0.0019*,† [0.0018, 0.0019]	yes	—	—
	dec. k=4	0.009*,† [0.007, 0.011]	yes	−0.0023*,† [−0.0023, −0.0022]	yes	—	—
LLaDA Base TS	uniform	−0.001 [−0.003, 0.002]	—	0.0017† [0.0016, 0.0018]	—	—	—
	dec. k=4	0.008† [0.004, 0.012]	—	−0.0010† [−0.0011, −0.0008]	—	—	—
Dream Base TS	uniform	0.002 [−0.001, 0.004]	—	0.0041† [0.0039, 0.0043]	—	—	—
	dec. k=4	0.012† [0.008, 0.015]	—	0.0010† [0.0008, 0.0011]	—	—	—
MDLM-110M TS	uniform	0.012† [0.009, 0.015]	—	0.0048† [0.0047, 0.0050]	—	—	—
	dec. k=4	0.005† [0.001, 0.009]	—	0.0004† [0.0001, 0.0005]	—	—	—

Backup: β_z with intervals

Without and with state effects; a diagnostic, no Holm family

Arm	Masks	β_z, no FE	β_z, FE
LLaDA Base	uniform	—	—
	dec. k=4	—	—
Dream Base	uniform	—	—
	dec. k=4	—	—
LLaDA-I GSM8K	uniform	—	—
	dec. k=4	—	—
LLaDA-I MBPP	uniform	—	—
	dec. k=4	—	—
Dream-I GSM8K	uniform	—	—
	dec. k=4	—	—
Dream-I MBPP	uniform	—	—
	dec. k=4	—	—
MDLM-110M	uniform	—	—
	dec. k=4	—	—
LLaDA Base TS	uniform	—	—
	dec. k=4	—	—
Dream Base TS	uniform	—	—
	dec. k=4	—	—
MDLM-110M TS	uniform	—	—
	dec. k=4	—	—

Backup: T8 controls, every arm

E1 and E2 without the control and with it; ‡: paired difference excludes 0

Arm	Masks	E1	+pos.	+type	+temp.	E2	+pos.	+type	+temp.
LLaDA Base	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
Dream Base	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
LLaDA-I GSM8K	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
LLaDA-I MBPP	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
Dream-I GSM8K	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
Dream-I MBPP	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
MDLM-110M	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
LLaDA Base TS	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
Dream Base TS	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—
MDLM-110M TS	uniform	—	—	—	—	—	—	—	—
	dec. k=4	—	—	—	—	—	—	—	—

Backup: C1 robustness

None of these is in a Holm family

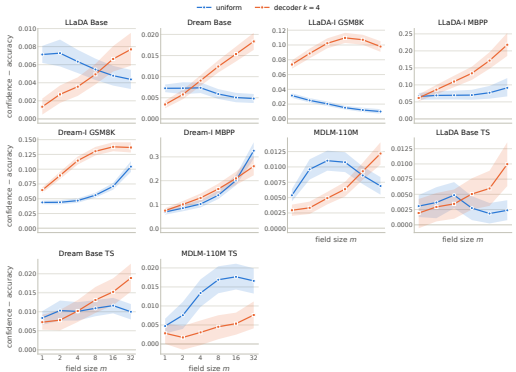
Arm	Masks	E1	E1 clean	E2, 5 kn.	E2, 9 kn.	E2, 13 kn.	E2 clean	E2 add.
LLaDA Base	uniform	−0.003†	—	—	0.001†	—	—	—
	dec. k=4	0.006†	—	—	0.004†	—	—	—
Dream Base	uniform	−0.002†	—	—	0.001†	—	—	—
	dec. k=4	0.015†	—	—	0.003†	—	—	—
LLaDA-I GSM8K	uniform	−0.022†	—	—	−0.003†	—	—	—
	dec. k=4	0.024†	—	—	0.021†	—	—	—
LLaDA-I MBPP	uniform	0.025†	—	—	0.013†	—	—	—
	dec. k=4	0.156†	—	—	0.036†	—	—	—
Dream-I GSM8K	uniform	0.061†	—	—	0.021†	—	—	—
	dec. k=4	0.072†	—	—	0.038†	—	—	—
Dream-I MBPP	uniform	0.257†	—	—	0.176†	—	—	—
	dec. k=4	0.186†	—	—	0.076†	—	—	—
MDLM-110M	uniform	0.002†	—	—	0.002†	—	—	—
	dec. k=4	0.009†	—	—	−0.002†	—	—	—
LLaDA Base TS	uniform	−0.001	—	—	0.002†	—	—	—
	dec. k=4	0.008†	—	—	−0.001†	—	—	—
Dream Base TS	uniform	0.002	—	—	0.004†	—	—	—
	dec. k=4	0.012†	—	—	0.001†	—	—	—
MDLM-110M TS	uniform	0.012†	—	—	0.005†	—	—	—
	dec. k=4	0.005†	—	—	0.000†	—	—	—

Masks	Effect	fp16	4-bit	Difference
—	—	—	—	—

Reading. A column equal to the plain E1 or E2: the claim does not depend on the knots or on contaminated text. E2 uses a calibrator, E1 does not.

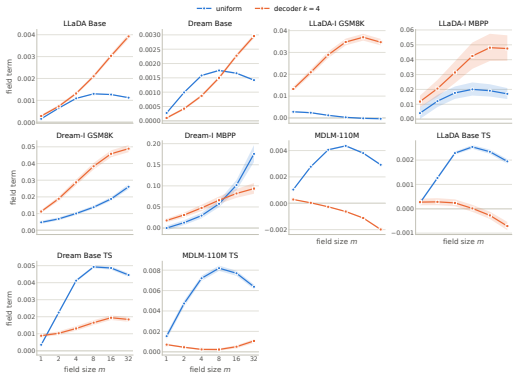
Backup: $\Delta(m)$ in every arm

Blue: uniform states; orange: decoder k=4; 95% bands



Backup: the field term in every arm

Blue: uniform states; orange: decoder k=4; 95% bands



Backup: E3 in every arm

Type effect at equal confidence against matched uniform states and the reweighted uniform grid

Arm	k	Matched uniform	p	Uniform grid	p
LLaDA Base	—	—	—	—	—
Dream Base	—	—	—	—	—
LLaDA-I GSM8K	—	—	—	—	—
LLaDA-I MBPP	—	—	—	—	—
Dream-I GSM8K	—	—	—	—	—
Dream-I MBPP	—	—	—	—	—
MDLM-110M	—	—	—	—	—

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